

UNCOVERING A HIDDEN WORLD: ALGAL–FUNGAL RELATIONSHIPS IN COOK’S BAY, MO’OREA

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Abstract. Marine fungi remain one of the least understood components of marine ecosystems. This study provides an initial characterization of fungal communities associated with macroalgae in the fringing reef of Cook’s Bay, Mo’orea, a region previously uncharacterized for marine fungi. Algal samples from nine host species were collected and cultured, and any fungal isolates were sub-cultured to the third generation and grouped into morphospecies based on colony morphology, growth patterns, spore characteristics, and hyphal septation. In total, 44 fungal species were identified from 78 third-generation isolates. Fungal community composition did not differ significantly among algal host species based on PERMANOVA, although host identity explained a small proportion of variation ($R^2 = 0.12$). Most isolates exhibited septate hyphae, suggesting a predominance of Ascomycota and Basidiomycota, though taxonomic identities were not genetically confirmed. A complementary decomposition experiment evaluated fungal effects on algal mass loss using *Padina boryana* under sterilized and non-sterilized conditions. All samples showed either no change or mass loss, but decomposition was significantly greater in non-sterilized treatments, indicating that surface-associated microbial communities contribute to algal degradation. Together, these results establish a baseline for marine fungal diversity in Cook’s Bay and highlight the complexity of algal–fungal interactions, providing a foundation for future studies incorporating broader host sampling and molecular approaches.

Key words: *marine fungi, mushrooms, macroalgae, island, temperature, Padina Boryana, French Polynesia, host impact, diversity, species relationships, Ascomycota, Basidiomycota, mycology, decomposition, community composition, fungal ecology*

INTRODUCTION

Marine mycology is a field of study that has emerged within the last 150 years (Pang *et al.* 2023). Although significant progress in the field has been made, one of the central challenges in the marine mycology world is narrowing down a definition of what exactly constitutes a marine fungus (Pang *et al.* 2016). Many fungi that are found in the ocean are also found on land, which makes it challenging to discern whether marine fungi have an ecosystem function in marine environments or if spores are in a resting stage waiting to wash up onto a terrestrial environment to fruit (Kohlmeyer and Kohlmeyer 1979). Fungi on land typically grow as a network of hyphae known as a mycelium, with the fruiting body only emerging for reproduction. Yet virtually every ecological aspect differs in a marine environment, which leads to the central question: what does it mean to be a fungus in the ocean?

Pang *et al.* (2016) suggests a broad and generally accepted definition of marine fungi, “any fungus that is recovered repeatedly from marine habitats and: 1) is able to grow and/or sporulate (on substrata) in marine environments; 2) forms symbiotic relationships with other marine organisms; or 3) is shown to adapt and evolve at the genetic level or be metabolically active in marine environments”. With this definition in mind, marine fungi have been found in almost every marine environment, from the bottom

of the deep sea to intertidal coastlines (Amend *et al.* 2019). Furthermore, Marine fungi have been cultured from a plethora of hosts including submerged wood, sessile and mobile invertebrates, marine mammals, algae, and sediments (Amend *et al.* 2019). As of September 2025, there have been 2253 species identified under 12 Phyla, 36 Classes, 117 Orders, 301 Families, and 881 Genera, with the vast majority belonging to the phyla Ascomycota and Basidiomycota (Jones *et al.* 2019).

Marine fungi play a biogeochemical role of cycling nutrients like carbon, nitrogen, and phosphorus (Peng *et al.* 2024), but little is known about what exactly marine fungi are doing and how they are affecting marine ecosystems on a large scale. Mo’orea, an island in the Society Island chain, is especially sensitive to the changing temperatures of atmospheres and oceans, as islands are disproportionately impacted by climate change (Martin *et al.* 2018). Macroalgae, multicellular autotrophs that respond strongly to changing temperature (Choudhary 2021), are well characterized in Mo’orea with 34 species described as commonly found (Hinkle *et al.* 1999). Macroalgae in Mo’orea may serve as key hosts influencing fungal community composition. Macroalgae blooms and tides of large overgrowth are correlated to heightened temperatures and levels of CO₂, but in some cases, rising temperatures can hinder growth and success of macroalgae as they are exposed to levels above their

thermal limit (Ji & Gao 2020). It is not clear exactly how changing temperature will affect algae and specifically their fungal communities.

Recent studies have characterized marine fungal communities in the Central Pacific (Peng *et al.* 2024), however, comparable work has not been conducted in the Society Islands. This study aims to investigate the relationships between macroalgae inhabiting the fringing reef of Cook's Bay and the fungal communities growing on and within these hosts. Specifically, this research addresses two primary objectives. The first objective is to characterize the fungal taxa associated with macroalgae in Cook's Bay and to assess how algal host identity influences fungal community composition and diversity. The second objective is to determine whether fungi contribute to algal decomposition and, if so, to quantify the extent of decomposition and identify the conditions under which it occurs. I hypothesize that algal host identity plays a significant role in the structure of associated fungal communities, given the substantial diversity among algal species in size, blade and branching morphology, and taxonomic group (red, green, and brown algae). This morphological and functional diversity among hosts is expected to correspond to variation in the composition of associated fungal taxa. I further hypothesize that fungi do contribute to the decomposition of their algal hosts.

MATERIALS AND METHODS

Study site and field collections

This study was conducted in the fringing reef directly off Gump Research Station in Cook's Bay, Moorea. The entirety of the study occurred between the months of October and November 2025.



FIG. 1. Site sampled in this study (red dot).

Set up

The type of media used for fungi growth was PDA (Potato Dextrose Agar), and SA (Seawater Agar) (Table 4; Appendix E). All plates, tweezers, scalpels, and samples were handled with lab gloves, and a mask was worn during the entirety of the sub-culturing process to prevent inhalation of spores.

Objective 1 methods

To determine how algal hosts impact fungal communities, algae samples had to be collected, fungi had to be cultured from these samples, and finally, the fungi had to be identified.

Collecting algae was the first step. All algae samples were collected by snorkeling in the fringing reef right off the Gump Station. The collection side started at the dock and ended at the sandy beach in front of two water bungalows. Algae were collected at random until an adequate amount of the algae species was collected, about 7 individuals at one time, and then placed in an opaque, half gallon sized sterile container, and immediately transported to the laboratory and plated within 8 hours (Ettinger *et al.* 2021). Nine species of algae ended up being collected, because they appeared to be the most prevalent. The process of plating each algae sample involved 3 main steps:

1. Pieces of algae were cut into roughly 0.5 cm sized portions using an ethanol-lamp sterilized dissection scalpel. This typically was done in a square shape if the algae sample had a large enough surface area, like *Padina boryana*, but for algae species like *Amphiroa fragilissima*, which has a thin, tubular formation, the sample was simply cut at ~ 0.5 cm length.

2. Next, sterilized seawater was poured in three small, glass vials, and all samples were rinsed in these vials for thirty seconds each, three times (Cha *et al.* 2021). The seawater was replaced after each rinse.

3. Next, the collection of samples were split in half. Half of the samples were placed on paper towels in petri dishes to dry, and the other half were rinsed in 70% ethanol for thirty seconds, once, then thoroughly rinsed again with sterilized seawater three more times for thirty seconds each (Cha *et al.* 2021). Finally, this half was set out on paper towels in petri dishes to dry. This sterilization method was used to access fungi living on the surfaces of samples (non-sterilized), as well as fungi living inside of the samples (sterilized).

4. Lastly, each dried sample was then plated by using an ethanol lamp sterilized pair of tweezers, on a petri dish with PDA. 3-4 samples were placed on one petri dish to be conservative with materials, and labeled appropriately, using the created labeling system (Table 3; Appendix D)

After this plating process, all plates were evenly distributed among three different temperature incubators. The incubators were as follows: A at 27° C, B at 32° C, and C at 26° C. The incubators were made from large, 50-gallon black tubs that were filled with approximately 7000 mm of water. Then, an aquarium heater was submerged under the water. A gallon sized white tub was then placed floating on top of the heated water. An infrared thermometer was used to detect the ambient air temperature inside these tubs. All plates were checked every 1–2 days for mycelium growth. If there was growth, it was sub-cultured by using an ethanol-lamp-sterilized dissection scalpel to cut a .05 cm chunk from the edge of the mycelium (where the mycelium was actively growing and expanding). This chunk was placed on a new PDA filled petri dish, labeled, and placed in the same temperature incubator as its parent. This process was repeated to the third generation, meaning the first generation (algae sample) was sub-cultured once to the second generation, and then finally the second generation was sub-cultured to the third generation. At the beginning of each new plate, including the algae sample plates, photos using an iPhone 15 were taken of each plate to be able to refer to. As soon as the plating process for a petri dish was completed, the petri dish was closed and wrapped in two layers of either electrical tape (in absence of parafilm) or in two layers of parafilm. This entire sub-culturing process was done to obtain a single isolate. A single isolate is defined by having sub-cultured the organism two times, each with consistent morphology (Ettinger and Eisen 2020). Each original plate was observed for a minimum of four weeks (Bovio *et al.* 2016), and at most for 37 days, due to different sampling and plating dates.

Control plates were included in each incubator from the first plating date and at the last sub-culturing date. These control plates included PDA with no algae samples or mycelium plated. These were to observe any non-marine fungi spores that could possibly grow on PDA in the environments in which all other plates were being stored.

On 17/10/25 and 20/10/25, a total of 125 third generation individuals were plated. These were then stored in three-gallon sized, opaque tubs. For the first 8 days, pictures of each individual were taken with a ruler in the frame, again for future reference in case growth rate was a factor that would be needed to differentiate potential species. iPhone pictures of each individual were taken along with a hand lens and a light microscope at least once throughout the 17 days. The hand lens captured texture, and the microscope showed spores, mycelium growth patterns, and what the mycelium looked like at a microscopic level. I

most often used the 40x magnification and 100x magnification.

After 17 days and after accounting for control plates and contamination, 78 third generation individuals were ready to be analyzed for identification. The identification process used morphological traits to create morphospecies, and this involved grouping individuals by color, texture, growth pattern, spore morphology, and hyphae septation. Individuals were first grouped by color, then by texture, and then by growth pattern. Finally, microscope photos of spores and mycelium were used to either confirm or deny the potential species groupings.

Objective 2 Methods

To test if fungi have decomposition impacts on algae, 18 *Padina boryana* algae were collected using the same methods as mentioned above: snorkeled from the dock to the sandy beach in front of two water bungalows, collected at random and placed immediately in a sterilized gallon tub and brought to the lab. Fifteen thin, rectangular slices taken from leaflets of different individuals with a height of about .05 cm, and width spanning the length of the algae leaflet (varied), were cut and then cut in half, for a total of 30 slices, all with a height around .05 cm. All 30 slices were washed thoroughly with sterilized seawater twice, for thirty seconds each, using the same rinsing method described for the first objective.

Then, 15 of those samples were set aside on paper towels in petri dishes to dry. The other 15 were immersed in 70% ethanol for 30 seconds, then thoroughly washed it with sterilized seawater two times for thirty seconds each, then also set on paper towels to dry. After 3 days of drying, each sample was weighed using an analytical balance scale with a precision of 0.001 g. Before plating each sample on their own SA petri dish, a small chunk of fungi-colonized agar grown from a previous *Padina boryana* sample, with a size of around .25cm³, was cut (using an ethanol lamp sterilized scalpel) and plated on the SA plate with the mycelium facing upwards. Then, one end of the dried *Padina boryana* sample was carefully placed on top of the mycelium chunk, using an ethanol-lamp sterilized sample. Next, another chunk of mycelium was cut and placed mycelium face down onto that same end of the *Padina boryana* algae. This method, suggested by M. Garbelotto (University of California, Berkeley; personal communication) was used to sandwich the dried *Padina boryana* slices as a sort of inoculation. This was repeated for all 30 samples. 15 plates of the non sterilized samples and 15 plates of the ethanol sterilized samples were labeled with A, B, and C (10 in total for each grouping). A

plates were set in a covered black incubator tub inside the laboratory at 27° C, B plates were placed in a covered black incubator tub at 32° C, and C plates were set in a covered black incubator tub in the water tables at 26° C.

Control plates with slices of *Padina boryana* both ethanol sterilized and non-ethanol sterilized without any fungal inoculation were placed in A and C incubators. This was done to understand if weight change did occur, how much was due to the fungus and how much was due to outside factors. Each plate was monitored daily for contamination.

At the end of 13 days, the *Padina boryana* slice from each plate was separated from the mycelium chunks, thoroughly rinsed with sterilized seawater three times, dried overnight, and then weighed in the analytical balance scale.

Statistical analyses

All analyses were conducted in R studio (version 2025.09.0+387) using the packages tidyverse, vegan, and stats. Fungal community composition was analyzed using a plate-level species abundance matrix, and Bray–Curtis dissimilarity was used to quantify differences among samples. Differences in fungal community composition among algal host species were tested using PERMANOVA (adonis2, 999 permutations), with effect size reported as R^2 .

For the decomposition experiment, mass change was calculated as final minus initial mass for each plate. A one-sample t-test was used to test whether mean mass change differed from zero, and a Welch two-sample t-test was used to compare mass change between ethanol-sterilized and non-sterilized treatments. Statistical significance was assessed at $\alpha = 0.05$.

RESULTS

Objective 1 results

After conducting this observational experiment for 37 days, a total of 78 fungus individuals were collected. From there, 44 unique species were identified from morphological traits and labeled each species with letters. See Appendix C for a visual description of each morphological grouping.

The species richness composition seen in Fig. 2 shows how different algae can support a different number of species. *Turbinaria ornata* had 14 cases where individual plates from separate species were grown, at least once, from its algae. Fig. 2 does not say anything about what these fungus individuals are, but it does show a distinct range in species richness.

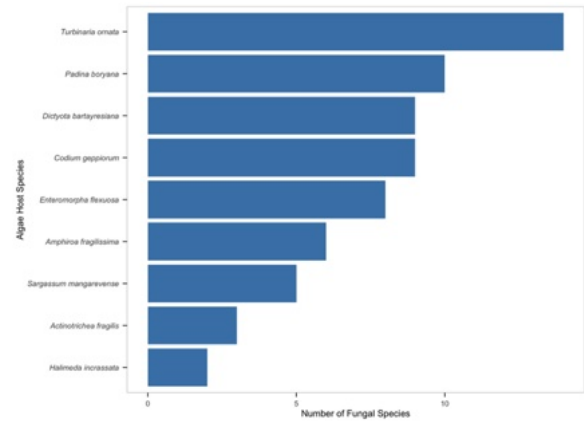


FIG. 2. Species richness distribution across 9 sampled algal hosts.

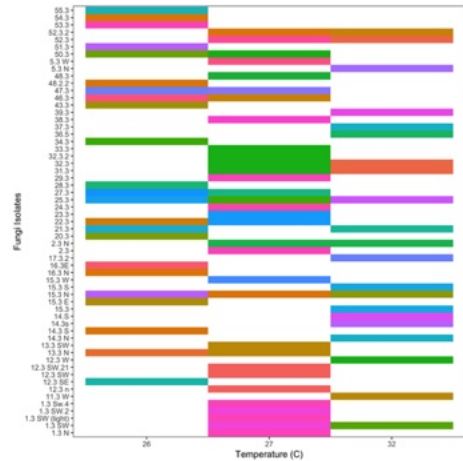


FIG. 3. Distribution of fungal isolates and their respective species across the temperature gradient.

Displaying the distribution of fungal species across algal hosts by temperature (Fig. 3) provides important contextual information for interpreting host–fungus associations. Although this figure does not directly test the primary research question, it illustrates how fungal recovery varies across incubation temperatures, highlighting differences in community structure and host occupancy that may contribute to the observed patterns in fungal diversity. Plates were separated by temperature to primarily provide the opportunity for as many different species as possible.

The next step that was taken to uncover a relationship pattern between algae and fungi was to group species by their morphospecies, instead of their species label. Fig. 3 depicts species groupings by color

and Fig. 4 by spore traits, and it is difficult to visually see any sort of pattern from these groupings and their algal hosts in either of these plots.

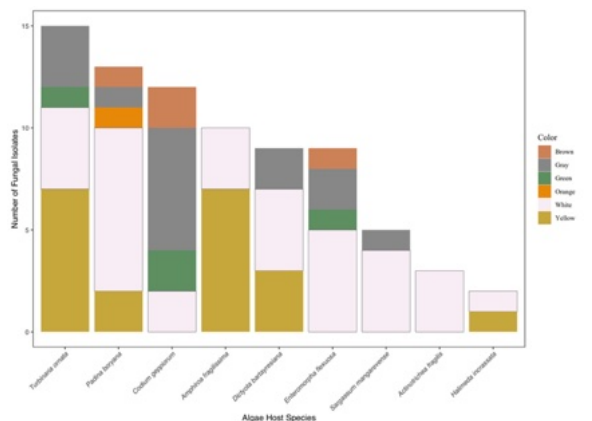


FIG. 3. Distribution of fungal species grouped by color.

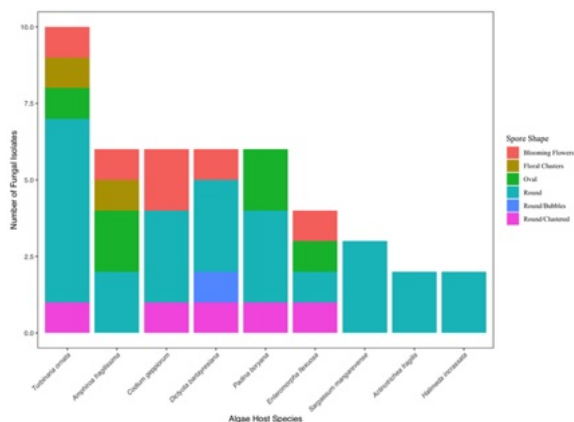


FIG. 4. Distribution of fungal species grouped by spore shape.

Fig. 5 displays an association matrix, showing the association between fungal species across algal hosts. On the x axis are the nine algae hosts and on the y axis are the 44 algae fungal species. Each blue bar shows the presence of the fungus species on a particular alga. The plot is sorted from highest association to lowest from left to right and from top to bottom. The color key on the right shows the replicability of the fungus grown. The darker the blue, the more plates of fungi showed the same result. A PERMANOVA (Bray Curtis) test was used to statistically test association. The null hypothesis that the fungal community composition does not differ among algal host species was tested against the alternative hypothesis, that fungal community composition differs among algal host species, and based on this PERMANOVA test, a p value of .062

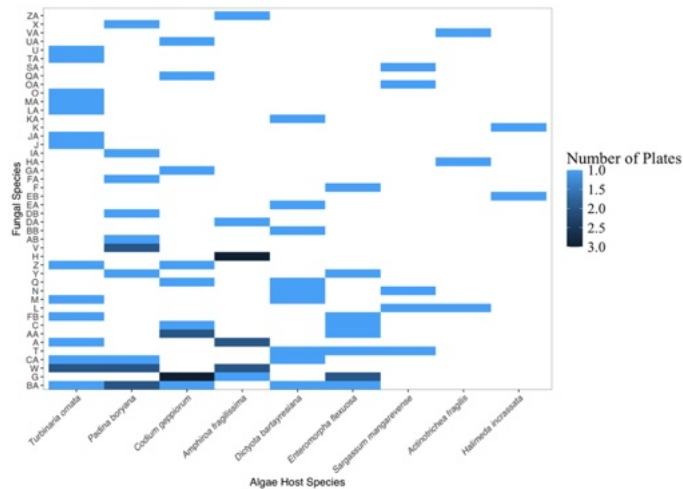


FIG. 5. Fungal species and algae host association matrix.

was reached. Another value obtained from this test is the R^2 , in this case being 0.12, which represents the fraction of total variation in Bray–Curtis community dissimilarities explained by algal host identity.

Objective 2 results

Comparison in weight change between control plates and treatment plates was needed to understand whether decomposition occurred at a statistically significant rate. Fig. 6 shows that decomposition did in fact occur, and this was confirmed by using a one-sided, Welch, two-sample t test. We tested the null hypothesis that treatment plates were not lesser in mean weight change compared to control plates, against the alternative that treatment plates exhibited greater weight loss. This resulted in a p value of 0.036.

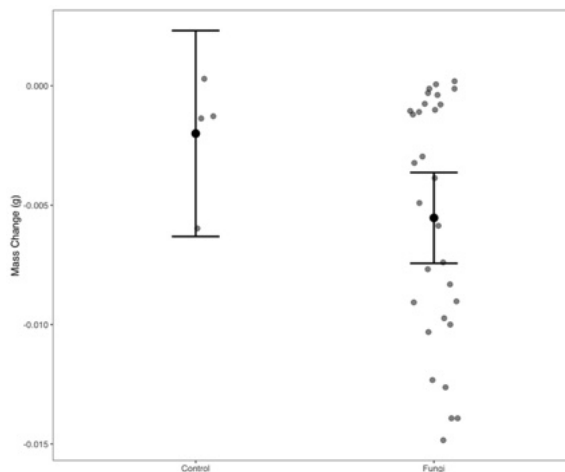


FIG. 6. Treatment vs. control plates mass change over 13 days.

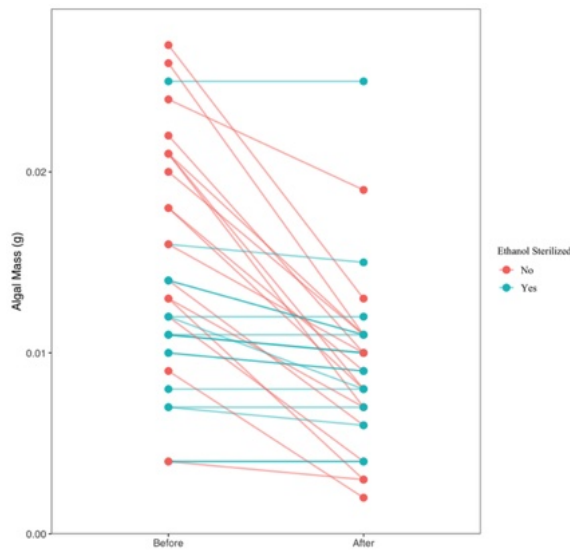


FIG. 7. Mass change over 13 days by ethanol sterilization.

Because this fungus decomposition was confirmed, ethanol sterilization was then looked at as a variable from the linear regression plotted because it seemed like there was a distinct visual pattern between samples that had been sterilized using ethanol, and those that had not, as seen in Fig 7. Using a two-sided Welch, two sample t-test, with the null hypothesis being that the mean algal weight change does not differ between ethanol-sterilized and non-sterilized plates. From this test, $P = 3.5 \times 10^{-8}$ was extracted. Non-sterilized plates exhibited substantially greater algal mass loss (mean $\pm$ SD = -0.00924 ± 0.00368 g) than ethanol-sterilized plates (-0.00100 ± 0.00122 g), indicating that ethanol treatment strongly reduced decomposition.

DISCUSSION

Research question and hypothesis overview

This study investigated the relationships between macroalgae in the fringing reef of Cook's Bay and their associated fungal communities. Specifically, it examined how algal host identity influences fungal community composition and diversity, and whether fungi contribute to algal decomposition under varying conditions. Given the morphological and taxonomic diversity of macroalgae, host identity was expected to structure associated fungal communities. Additionally, based on the known role of fungi in terrestrial systems, fungal involvement in algal decomposition was anticipated.

Summary of significant results

After 37 days from beginning to end, 44 unique fungal species were identified from 78 plates across 9 different species of algae. A statistically significant association was not found between algal host and fungal species. No pattern was found when examining fungal species grouped by color or spore morphology. Decomposition occurred on *Padina boryana* samples that had been inoculated with fungus, and the amount of decomposition was strongly impacted by ethanol sterilization.

Objective 1 summary

The sampling study indicates that the algae inhabiting the fringing reef of Cook's Bay support an exceptionally diverse fungal community. Of the 78 isolates, 21 were grown from incubation at 32°C, 33 at 27°C, and 24 at 26°C. All algal hosts were represented at least once, with the most prevalent being *Turbinaria ornata*, which supported 15 individuals; *Padina boryana*, which supported 13 fungi; and *Codium geppiorum*, which supported 12 fungi (Fig. 2). The three fungal species with the highest number of isolates were species W, BA, and G, with six individuals each.

These distribution and species richness results are important, but there is a complication in these results. The complication involves the number of each algal sample used in this study. A total of 227 samples were initially plated; however, there were only 128 third-generation plates. This means that during the process, 109 plates were either contaminated or showed no visible growth. Additionally, the distribution of algae among plates was not even. Of the original 227 plates, 44 were from *Turbinaria ornata* samples, 30 were from *Amphiroa fragilissima*, and 24 were from *Halimeda incrassata*. Six plates were from *Acanthophora spicifera* samples, which eventually had to be abandoned due to insufficient collection of this alga. The lowest number of plates among the analyzed algae was 18, from a filamentous alga that could not be identified. Although this is still a substantial number of samples for each algal species with, it is important to note that the wide range of plate numbers can skew results, especially for measurements such as species richness (Fig. 2). It was difficult to achieve equal sampling across algal hosts due to high contamination rates in this study.

Still, these findings without a doubt suggest that fungal communities across the sampled algae are highly complex. The presence of 44 species across 9 different algal hosts demonstrates that algae can support multiple fungal species simultaneously and that these fungi coexist within a shared ecological

space. Another study sampling marine fungi in the South Pacific Ocean found a majority of Basidiomycota, Ascomycota, and fungi incertae sedis (Peng *et al.* 2024). Although it is beyond the scope of this study to conclusively identify individuals to the phylum level, it is probable that many of the individuals grown in this study belong to Basidiomycota and Ascomycota. Of the third-generation isolates, 41 displayed septate hyphae, 29 were inconclusive, and 8 showed aseptate hyphae. Chytridiomycota generally have aseptate hyphae, while Ascomycota and Basidiomycota display septate hyphae (Naranjo-Ortiz and Gabaldón 2019). While again, this is not conclusive, it provides additional context regarding the fungal groups examined in this study.

In the association matrix (Fig. 5), we failed to reject the null hypothesis that fungal community composition does not differ among algal host species. The PERMANOVA p-value of 0.064 indicates that the observed differences in community composition are not statistically distinguishable from random variation at $\alpha = 0.05$. In other words, the centroids of fungal community composition (based on Bray–Curtis dissimilarity) do not differ significantly among algal hosts, and any observed differences are no greater than expected by chance. Importantly, failure to reject the null hypothesis does not imply that algal host identity has no effect, but instead that there is insufficient evidence to support a statistically significant effect given the observed variability. The R^2 value of 0.12 indicates that algal host identity explains approximately 12% of the variation in fungal community composition, while the remaining 88% of variation is associated with other factors such as environmental conditions, fungal competition and colonization, or plate-level variability. Overall, these results suggest that algal host identity plays a meaningful but non-dominant role in structuring fungal communities.

Objective 1 implications

The ecological implications of this portion of the project are those of standard natural history observations. These findings establish baseline biodiversity documentation of the species currently present in Cook's Bay, Mo'orea, a region that is entirely uncharacterized with respect to marine fungi. This baseline is essential for current and future understanding of marine fungi, as it can be used to compare ecological changes over time. Furthermore, the high uniqueness among fungal individuals and species reflects a strong capacity for diverse fungi to thrive on algae and within this marine environment.

Objective 2 Summary

The experimental portion of this study measured weight change in 30 *Padina boryana* samples inoculated with a fungus before and after 13 days of incubation. The results from the one-sided, Welch, two-sample t test on control plates vs treatments plates (Fig. 6) highlight the decomposition potential of fungal activity on algal hosts. The p value of 0.036 and means that we can reject the null hypothesis and that there is statistically significant decomposition from *Padina boryana* inoculated with this specific fungus.

Half of the samples were not sterilized with ethanol, while the other half were sterilized, effectively eliminating surface-dwelling fungi and bacteria. Weight loss was far greater in the non-sterilized samples, whereas sterilized samples exhibited only minimal changes. With a p value of 3.5×10^{-8} obtained by using a two-sided Welch, two sample test, there is overwhelming evidence that ethanol sterilization impacts the rate of degradation of this fungus on *Padina boryana*, and that we can reject the null hypothesis.

This is an outcome that was unexpected and somewhat counterintuitive. One interpretation is that surface fungal and/or bacterial communities may provide a foundation that facilitates colonization by the inoculated fungus. Alternatively, these surface microbes may not only support colonization but also actively contribute to the decomposition process themselves. In either case, the results suggest that the naturally occurring microbial layer on algal surfaces plays a key role in algal decomposition.

Objective 2 implications

This experiment has ecological relevance and potential environmental implications, particularly for island ecosystems such as Mo'orea. The experiment was designed to test whether fungi decompose algae and, if so, what factors influence decomposition. *Padina boryana* was selected as the focal species due to its particularly high abundance in Cook's Bay during the sampling season. During this period, the reef system in Cook's Bay was in an algal-dominated phase, with algae such as *Padina boryana* overtaking coral cover. Algal blooms often develop in warm-water boom-and-bust cycles, and the unusually high levels of *Padina boryana* observed in this study area prompted further inquiry: What drives the decline phase of this cycle? Could fungal activity contribute to algal turnover? Does elevated temperature trigger increased decomposition, and if so, could this process eventually be applied in the context of managing

future blooms that are likely to become more common as ocean temperatures rise?

Overall project limitations

This project included multiple components and was therefore subject to several limitations. The most significant constraints were time and access to materials. Although efforts were made to accommodate slow-growing fungi, many plates could not be sub-cultured a second time, requiring some identifications to be based on the initial subculture or original plate. Previous studies have recommended three subcultures per isolate to increase identification accuracy (Ettinger & Eisen 2020), but this was not feasible and may affect the reliability of species identifications.

Material limitations also affected both studies. Only one growth medium was used (SA was used in the second study, but to inhibit growth), petri dishes were limited, and some essential growth ingredients were unavailable. As a result, multiple isolates sometimes had to be plated together, increasing the possibility of interactions between fungi. While this likely did not substantially impact results, it would be ideal to culture one fungal isolate per plate to minimize interactions that may influence growth. Additionally, because materials were unavailable for a third control condition in the experimental study, temperature treatment B lacked a control plate, reducing comparative robustness.

Most studies related to culturing marine fungi incorporate antibiotics to suppress bacterial growth and facilitate fungal isolation (Rämä *et al.* 2014). While a first-aid antibiotic ointment was used in some test plates, results were inconsistent, and this method was abandoned due to difficulty integrating the ointment into the media. Surprisingly, many fungi still grew successfully without antibiotics; however, a substantial number of plates were overrun by bacteria, preventing fungal growth.

Another limitation was the inconsistent temperature gradient. Incubation was conducted using a water incubator at 32°C, room temperature at 27°C, and outdoor water tables at approximately 26°C. Because colder incubation conditions were not possible, data from the latter two temperatures may be insufficient to reveal significant trends without larger sample sizes.

One of the largest limitations of this study was reliance on morphological characteristics for species identification in the absence of genetic sequencing. While morphology-based identification is valid and widely used, it cannot guarantee complete accuracy. Previous studies relied on morphological data but always in conjunction with DNA sequencing

(Chalearmsrimuang *et al.* 2019). For more conclusive and reliable results, genetic sequencing is recommended.

Lastly, it is important to note that the presence of a fungus on a petri dish does not necessarily prove that the fungus exists in the ocean as an active organism that reproduces and contributes to the broader ecosystem (Kohlmeyer and Kohlmeyer 1979). While it is unlikely that all 79 isolates were dormant in the marine environment, this possibility cannot be entirely excluded.

CONCLUSION

There is strong potential for this project to serve as a springboard into a more robust initiative, with a couple of considerations to keep in mind. For a future project, using a larger sample size of algae, as well as other types of hosts such as driftwood, seagrass, sediments, coconut husks, corals, and sea sponges, would add greater depth and improve the reliability of results. Although PDA is a well-known and successful medium for growing mushrooms, using other media would allow more possibilities for fungi that do not survive well on PDA to grow. Another future project could replicate the decomposition experiment with more time and more species of algae to get a more comprehensive understanding of what is happening in the relationship between fungi and algae. Finally, genetic sequencing would catalyze a project like this into a more reliable study.

Overall, this study provides an initial characterization of marine fungi living in the fringing reef of Cook's Bay, Mo'orea, including how they interact with algal hosts and with one another, and how they may contribute to environmental processes relevant to conservation. Taken as a whole, this work adds another piece to the largely unexplored and enigmatic puzzle of marine fungi.

AI STATEMENT

ChatGPT was used to assist with R coding, data visualization, and graphical formatting.

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APPENDIX A
Objective 1 data

TABLE 1.

Plate ID	Species	Algae host	Color	Simplified Color	Hyphal Septation	Physical Traits	Growth Pattern Specific	Growth Pattern	Spore Presence	Spore Shape Specific	Spore Shape	Temperature	Temperature C
A12.3 n	A	<i>Turbina ornata</i>	Mustard Yellow	Yellow	unclear	Stringy/Powdery	Uniform radial	Uniform radial	Present	Blooming flowers	Blooming Flowers	A	27
A12.3 SW. 21	A	<i>Amphiroa fragilissima</i>	Mustard Yellow	Yellow	unclear	Stringy/Powdery	Uniform radial	Uniform radial	Not present	NA	NA	A	27
A12.3 SW	A	<i>Amphiroa fragilissima</i>	Mustard Yellow	Yellow	unclear	Stringy/Powdery	Uniform radial	Uniform radial	Present	Blooming flowers	Blooming Flowers	A	27
A46.3	C	<i>Codium geppiorum</i>	Green	Green	unclear	Powdery/Fuzzy	Concentric rings	Concentric rings	Present	Round/Clustered	Round/Clustered	A	27
B52.3.2	C	<i>Enteromorpha flexuosa</i>	Green	Green	unclear	Powdery/Fuzzy	Concentric rings	Concentric rings	Present	Round/Clustered	Round/Clustered	B	32
A48.3	GA	<i>Codium geppiorum</i>	Gray	Gray	Septate	Fuzzy	Uniform radial	Uniform radial	Present	Round	Round	A	27
C50.3	F	<i>Enteromorpha flexuosa</i>	Gray/White	Gray	Aseptate	Thick	Uniform radial	Uniform radial	Present	Oval	Oval	C	26
A32.3	G	<i>Codium geppiorum</i>	Gray/White	Gray	Septate	Thin/Fuzzy	Bulbous	Bulbous	Not present	NA	NA	A	27
A32.3.2	G	<i>Codium geppiorum</i>	Gray/White	Gray	Septate	Thin/Fuzzy	Bulbous	Bulbous	Not present	NA	NA	A	27
A31.3	G	<i>Codium geppiorum</i>	Gray	Gray	Septate	Thin/Fuzzy	Bulbous	Bulbous	Not present	NA	NA	A	27
A50.3	G	<i>Enteromorpha flexuosa</i>	Gray/White	Gray	unclear	Thin/Fuzzy	Bulbous/Concentric rings	Concentric rings	Not present	NA	NA	A	27
A33.3	G	<i>Enteromorpha flexuosa</i>	White	White	unclear	Thin/Fuzzy	Bulbous	Bulbous	Not present	NA	NA	A	27
B12.3 W	G	<i>Amphiroa fragilissima</i>	White	White	unclear	Thin	Bulbous	Bulbous	Not present	NA	NA	B	32
A2.3	H	<i>Amphiroa fragilissima</i>	Yellow Hue/White	Yellow	Septate	Fluffy	Uniform radial	Uniform radial	Present	Round	Round	A	27
A2.3 N	H	<i>Amphiroa fragilissima</i>	Yellow Hue/White	Yellow	Septate	Fuzzy	Uniform radial	Uniform radial	Not present	NA	NA	A	27
B2.3 N	H	<i>Amphiroa fragilissima</i>	Yellow Hue/White	Yellow	Septate	Fuzzy	Uniform radial/Geometric	Uniform radial	Not present	NA	NA	B	32
A27.3	J	<i>Turbina ornata</i>	Yellow Hue	Yellow	Aseptate	Fluffy	Irregular radial, uniform	Irregular radial	Present	Round	Round	A	27
C12.3 SE	K	<i>Halimeda incrassata</i>	Yellow Hue	Yellow	Septate	Fluffy	Uniform radial/Vertical/Geometric	Uniform radial	Present	Round	Round	C	26
B37.3	L	<i>Actinotrichia fragilis</i>	White	White	Septate	Thin/Spike	Concentric rings	Concentric rings	Present	Round	Round	B	32
B14.3 N	L	<i>Sargassum mangroveense</i>	White	White	unclear	Thin/Spike	Bulbous/Concentric rings	Concentric rings	Present	Round	Round	B	32

B15 .3	M	<i>Dictyota bartayresiana</i>	Gray/White	Gray	unclear	Spikey, Cotton ball	Uniform radial/Vertical	Uniform radial	Present	Round	Round	B	32
B15 .3 S	M	<i>Turbinaria ornata</i>	Gray/White	Gray	unclear	Spikey, Cotton ball	Uniform radial/Vertical	Uniform radial	Present	Round	Round	B	32
A22 .3	N	<i>Dictyota bartayresiana</i>	White	White	Aseptate	Spikey	Uniform radial	Uniform radial	Present	Round	Round	A	27
A23 .3	N	<i>Sargassum mangroveense</i>	White	White	unclear	Spikey	Uniform radial	Uniform radial	Present	Round	Round	A	27
B36 .5	HA	<i>Actinotrichia fragilis</i>	White	White	Aseptate	Spikey	Uniform radial/Vertical	Uniform radial	Present	Round	Round	B	32
C27 .3	O	<i>Turbinaria ornata</i>	White/Pink Hue/Yellow Hue	White	Septate	Spikey	Uniform radial/Vertical	Uniform radial	Not present	NA	NA	C	26
A13 .3 N	CA	<i>Padina boryana</i>	Gray/White	Gray	Septate	Feathery/Spike y	Uniform radial	Uniform radial	Present	Oval/Round	Oval	A	27
B11 .3 W	CA	<i>Turbinaria ornata</i>	Gray/White	Gray	Septate	Feathery/Spike y	Uniform radial	Uniform radial	Not present	NA	NA	B	32
C47 .3	Q	<i>Codium geppiorum</i>	Gray/Brown	Gray	Septate	Thin, Cotton ball	Uniform radial/Vertical	Uniform radial	Present	Round	Round	C	26
B17 .3.2	Q	<i>Dictyota bartayresiana</i>	Gray/White	Gray	unclear	Thin, Cotton ball	Uniform radial/Vertical	Uniform radial	Not present	NA	NA	B	32
C51 .3	T	<i>Enteromorpha flexuosa</i>	White	White	septate	Spider web	Uniform radial	Uniform radial	Not present	NA	NA	C	26
B14 .3s	T	<i>Sargassum mangroveense</i>	White/Translucent/Orange	White	Septate	Thin/Flat	Uniform radial	Uniform radial	Present	Round	Round	B	32
C15 .3 N	T	<i>Dictyota bartayresiana</i>	White	White	Septate	Flat	Uniform radial	Uniform radial	Present	Round/Clustered	Round/Clustered	C	26
B14 .S	U	<i>Turbinaria ornata</i>	White/Translucent/Orange	White	Septate	Thin/Flat	Uniform radial	Uniform radial	Present	Round	Round	B	32
A1.3 SW.2	V	<i>Padina boryana</i>	White	White	Septate	Thin	Uniform radial	Uniform radial	Not present	NA	NA	A	27
A1.3 SW	V	<i>Padina boryana</i>	White	White	unclear	Thin	Uniform radial	Uniform radial	Present	Round	Round	A	27
A1.3 Sw.4	W	<i>Padina boryana</i>	Yellow Hue/White	Yellow	Septate	Fluffy	unclear	Uniform radial	Present	Round	Round	A	27
A1.3 N	W	<i>Turbinaria ornata</i>	Yellow Hue/White	Yellow	Septate	Fluffy	Uniform radial	Uniform radial	Not present	NA	NA	A	27
A29 .3	W	<i>Amphiroa fragilissima</i>	Yellow Hue/White	Yellow	Septate	Spikey	Uniform radial	Uniform radial	Present	Oval	Oval	A	27
C53 .3	W	<i>Turbinaria ornata</i>	Yellow Hue/White	Yellow	Septate	Fluffy/Spikey	Uniform radial	Uniform radial	Present	Round	Round	C	26
A2.3	W	<i>Amphiroa fragilissima</i>	Yellow Hue/White	Yellow	Septate	Spikey	Uniform radial	Uniform radial	Present	Oval	Oval	A	27
A1.3 SW (light)	W	<i>Padina boryana</i>	Yellow Hue/White	Yellow	unclear	Thin	Uniform radial	Uniform radial	Present	Round	Round	A	27
A24 .3	X	<i>Padina boryana</i>	White	White	unclear	Fuzzy/Spider web	Uniform radial	Uniform radial	Not present	NA	NA	A	27
A52 .3	Y	<i>Enteromorpha flexuosa</i>	White	White	Septate	Thick	Uniform radial	Uniform radial	Not present	NA	NA	A	27
C54 .3	Y	<i>Padina boryana</i>	White	White	Septate	Thick	Uniform radial	Uniform radial	Not present	NA	NA	C	26

B31 .3	AA	<i>Codium geppiorum</i>	Brown spores/Tran slucent mycelium	Brown	unclear	Powder y	Uniform radial/ Spotty	Uniform radial	Presen t	Bloomin g flowers	Bloom ing Flow ers	B	32
B32 .3	AA	<i>Codium geppiorum</i>	Brown spores/Tran slucent mycelium	Brown	unclear	Powder y	Uniform radial	Uniform radial	Presen t	Bloomin g flowers	Bloom ing Flow ers	B	32
B52 .3	AA	<i>Enterom orpha flexuosa</i>	Brown spores/Tran slucent mycelium	Brown	unclear	Fuzzy	Uniform radial/ Spotty	Uniform radial	Presen t	Bloomin g flowers	Bloom ing Flow ers	B	32
C14 .3 S	BA	<i>Turbina ria ornata</i>	White	White	Septate	Flat	Uniform radial	Uniform radial	Presen t	Round/ Clusture d lines	Roun d/Clu stere d	C	26
C16 .3 N	BA	<i>Padina boryana</i>	White	White	Septate	Flat	Uniform radial/Concent rate rings	Concent ric rings	Presen t	Oval and Bumpy	Oval	C	26
A15 .3 N	BA	<i>Dictyota bartayre siana</i>	White	White	Septate	Fluffy/ Black droplet s	Irregular radial/Concent rate ring/Geometri c	Irregula r radial	Not presen t	NA	NA	A	27
C48 .2.2	BA	<i>Codium geppiorum</i>	White	White	Septate	Fluffy/ Black droplet s	Concentric rings	Concent ric rings	Presen t	Round	Roun d	C	26
C54 .3	BA	<i>Padina boryana</i>	White	White	Septate	Thick/ Black droplet s	Uniform radial/Concent rate rings	Concent ric rings	Not presen t	NA	NA	C	26
A52 .3.2	BA	<i>Enterom orpha flexuosa</i>	White	White	Septate	Fluffy/ Black droplet s	Irregular radial/Geometr ic	Irregula r radial	Not presen t	NA	NA	A	27
C13 .3 N	AB	<i>Padina boryana</i>	White/Brow n	White	Septate	Cotton Ball	Uniform radial	Uniform radial	Not presen t	NA	NA	C	26
A13 .3 SW	CA	<i>Dictyota bartayre siana</i>	White	White	Septate	Feather y	Irregular radial	Irregula r radial	Presen t	Round	Roun d	A	27
C15 .3 E	DA	<i>Amphiro a fragiliss ima</i>	White	White	Aseptate	Fuzzy/ Spider web	Uniform radial	Uniform radial	Presen t	Floral Clusters	Flora l Clust ers	C	26
B15 .3 N	EA	<i>Dictyota bartayre siana</i>	Yellow	Yello w	Septate	Bubble s	Uniform radial	Uniform radial	Presen t	Round/ Bubbles	Roun d/Bu bbles	B	32
B1. 3 SW	FA	<i>Padina boryana</i>	Brown	Brown	Septate	Powder y	Spotty	Irregula r radial	Presen t	Round/F loral Clusters	Roun d/Clu stere d	B	32
A5. 3 W	Z	<i>Turbina ria ornata</i>	Green/Whit e	Green	Aseptate	Powder y	Uniform radial/Concent rate rings	Concent ric rings	Presen t	Elongate d (wheat- esk)	Oval	A	27
C46 .3	Z	<i>Codium geppiorum</i>	Green/Whit e	Green	unclear	Powder y	Uniform radial/Concent rate rings	Concent ric rings	Not presen t	NA	NA	C	26
C28 .3	IA	<i>Padina boryana</i>	Orange	Orang e	unclear	Slime	Irregular radial	Irregula r radial	Not presen t	NA	NA	C	26
B21 .3	JA	<i>Turbina ria ornata</i>	Yellow	Yello w	unclear	Fuzzy	Irregular radial/Vertical	Irregula r radial	Presen t	Floral Clusters	Flora l Clust ers	B	32
C55 .3	KA	<i>Dictyota bartayre siana</i>	Yellow	Yello w	unclear	Fuzzy	Uniform radial	Irregula r radial	Presen t	Bloomin g Flowers	Bloom ing Flow ers	C	26
C21 .3	LA	<i>Turbina ria ornata</i>	Yellow/Whi te	Yello w	unclear	Fluffy/ Slime	Irregular radial	Irregula r radial	Not presen t	NA	NA	C	26
C25 .3	MA	<i>Turbina ria ornata</i>	Yellow Hue	Yello w	unclear	Cluster ed	Irregular radial/Vertical	Irregula r radial	Presen t	Round	Roun d	C	26
A15 .3 W	OA	<i>Sargass um mangar evense</i>	White	White	Aseptate	Spikey	Uniform radial/Geometr ic	Uniform radial/G eometri c	Not presen t	NA	NA	A	27
A47 .3	QA	<i>Codium geppiorum</i>	Gray/White	Gray	Aseptate	Thin/F eathery	Uniform radial/Vertical	Uniform radial/G	Not presen t	NA	NA	A	27

B5.3 N	SA	<i>Sargassum mangroveense</i>	Gray	Gray	unclear	Thin/Fluffy	Uniform radial/Concentric rings	Uniform radial/Geometric	Not present	NA	NA	B	32
B25.3	TA	<i>Turbina ria ornata</i>	Dark Gray	Gray	Septate	Thin	Uniform radial	Uniform radial/Geometric	Not present	NA	NA	B	32
B39.3	UA	<i>Codium geppiorum</i>	White	White	unclear	Fuzzy	Irregular radial	Irregular radial	Not present	NA	NA	B	32
A38.3	VA	<i>Actinotrichia fragilis</i>	White	White	Septate	Sprigs	Uniform radial	Uniform radial/Geometric	Not present	NA	NA	A	27
C16.3E	ZA	<i>Amphiroa fragilissima</i>	White	White	Septate	Thick/Fluffy	Uniform radial/Concentric rings	Concentric rings	Present	Round	Round	C	26
C22.3	BB	<i>Dictyota bartayresiana</i>	Yellow Hue/White	Yellow	Septate	Thin/Flat	Uniform radial	Uniform radial/Geometric	Not present	NA	NA	C	26
C43.3	DB	<i>Padina boryana</i>	White	White	unclear	Feathery/Cotton ball	Irregular radial	Irregular radial	Not present	NA	NA	C	26
C20.3	EB	<i>Halimeda incrassata</i>	White	White	Septate	Powdery	Irregular radial/Vertical	Irregular radial	Present	Round	Round	C	26
A25.3	FB	<i>Turbina ria ornata</i>	White	White	Septate	Distinct horizontal lines	Uniform radial/Concentric rings	Concentric rings	Present	Round	Round	A	27
C34.3	FB	<i>Enteromorpha flexuosa</i>	White	White	unclear	Distinct horizontal lines	Uniform radial/Concentric rings	Concentric rings	Present	Round	Round	C	26

APPENDIX B

Decomposition Experiment Data

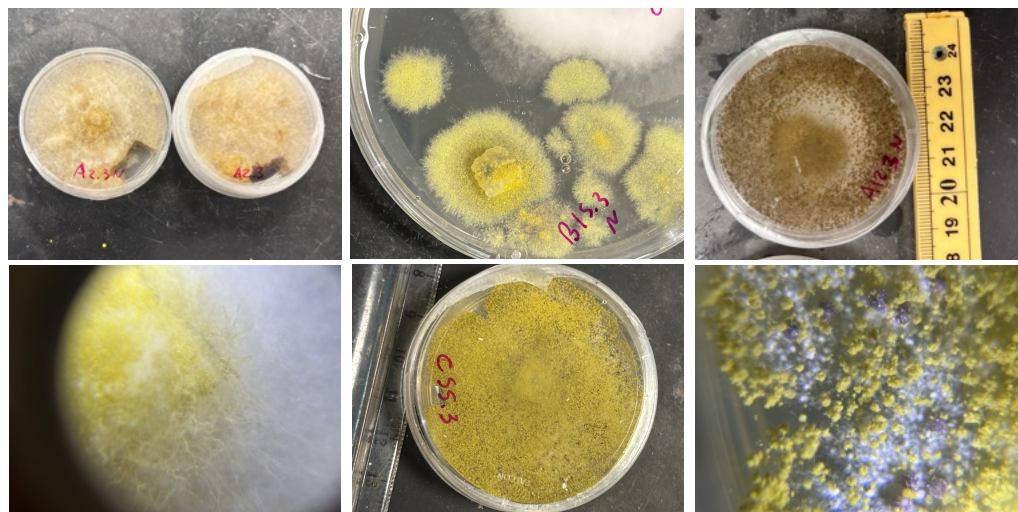
TABLE 2.

Plate_ID_1	Ethanol_Sterilized	Weight_1	Plate_ID_2	Weight_2	Temperature_Label	Temperature	Difference	Treatment
PA.1	No	0.026	PA.1	0.011	A	27	-0.015	Fungi
PA.2	No	0.021	PA.2	0.007	A	27	-0.014	Fungi
PA.3	No	0.022	PA.3	0.01	A	27	-0.012	Fungi
PA.4	No	0.014	PA.4	0.006	A	27	-0.008	Fungi
PA.5	No	0.021	PA.5	0.011	A	27	-0.01	Fungi
PB.1	No	0.024	PB.1	0.019	B	32	-0.005	Fungi
PB.2	No	0.013	PB.2	0.007	B	32	-0.006	Fungi
PB.3	No	0.013	PB.3	0.003	B	32	-0.01	Fungi
PB.4	No	0.018	PB.4	0.008	B	32	-0.01	Fungi
PB.5	No	0.027	PB.5	0.013	B	32	-0.014	Fungi
PC.1	No	0.009	PC.1	0.002	C	26	-0.007	Fungi
PC.2	No	0.02	PC.2	0.011	B	32	-0.009	Fungi
PC.3	No	0.012	PC.3	0.004	C	26	-0.008	Fungi
PC.4	No	0.018	PC.4	0.009	C	26	-0.009	Fungi
PC.5	No	0.021	PC.5	0.008	C	26	-0.013	Fungi
PA.1E	Yes	0.011	PA.1E	0.01	A	27	-0.001	Fungi
PA.2E	Yes	0.025	PA.2E	0.025	A	27	0	Fungi
PA.3E	Yes	0.008	PA.3E	0.008	A	27	0	Fungi
PA.4E	Yes	0.011	PA.4E	0.011	A	27	0	Fungi

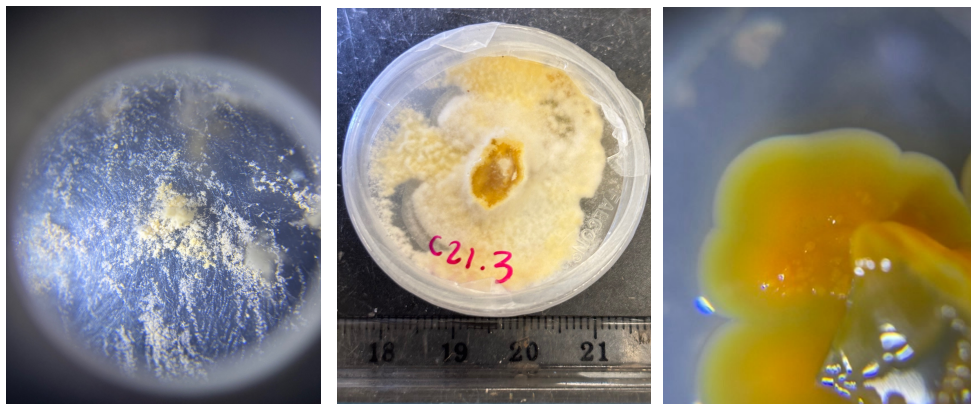
PA.5E	Yes	0.014	PA.5E	0.011	A	27	-0.003	Fungi
PB.1E	Yes	0.01	PB.1E	0.009	B	32	-0.001	Fungi
PB.2E	Yes	0.012	PB.2E	0.012	B	32	0	Fungi
PB.3E	Yes	0.011	PB.3E	0.01	B	32	-0.001	Fungi
PB.4E	Yes	0.01	PB.4E	0.009	B	32	-0.001	Fungi
PB.5E	Yes	0.011	PB.5E	0.01	B	32	-0.001	Fungi
PC.1E	Yes	0.016	PC.1E	0.015	C	26	-0.001	Fungi
PC.2E	Yes	0.007	PC.2E	0.007	C	26	0	Fungi
PC.3E	Yes	0.014	PC.3E	0.011	C	26	-0.003	Fungi
PC.4E	Yes	0.012	PC.4E	0.008	C	26	-0.004	Fungi
PC.5E	Yes	0.004	PC.5E	0.004	C	26	0	Fungi
C control E	Yes	0.007	C control E	0.006	C	26	-0.001	Control
C control	No	0.016	C control	0.01	C	26	-0.006	Control
A control E	Yes	0.004	A control E	0.004	A	27	0	Control
A control	No	0.004	A control	0.003	A	27	-0.001	Control

APPENDIX C

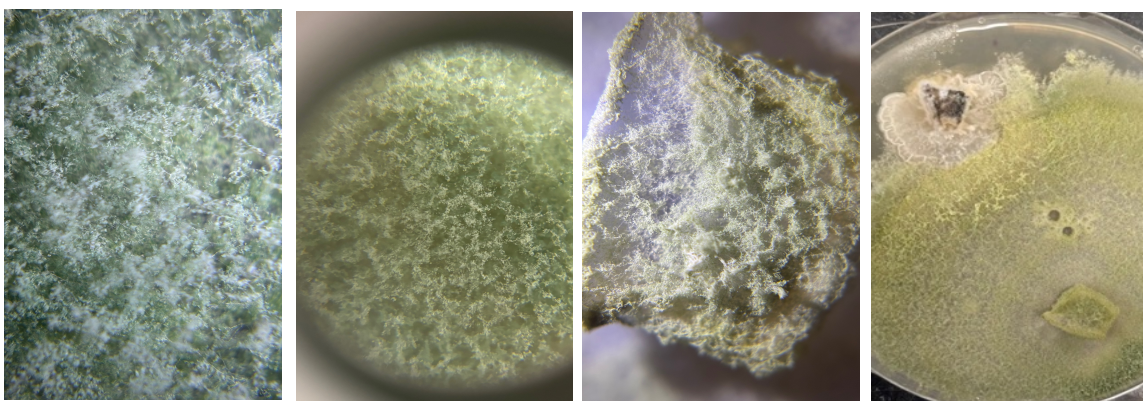
Morphospecies traits: color (C.1-C.4), texture (C.5-C.8) spore morphologies (C.9), hyphae septation (C.10)



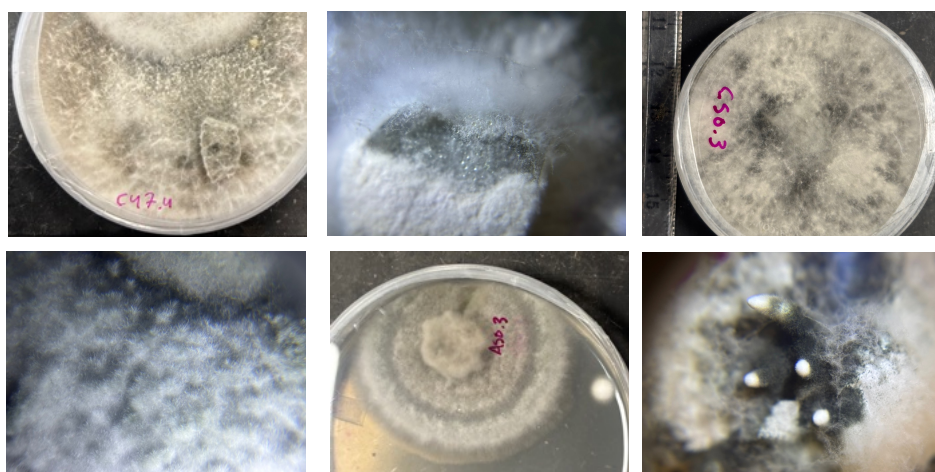
Appendix C.1 Yellow



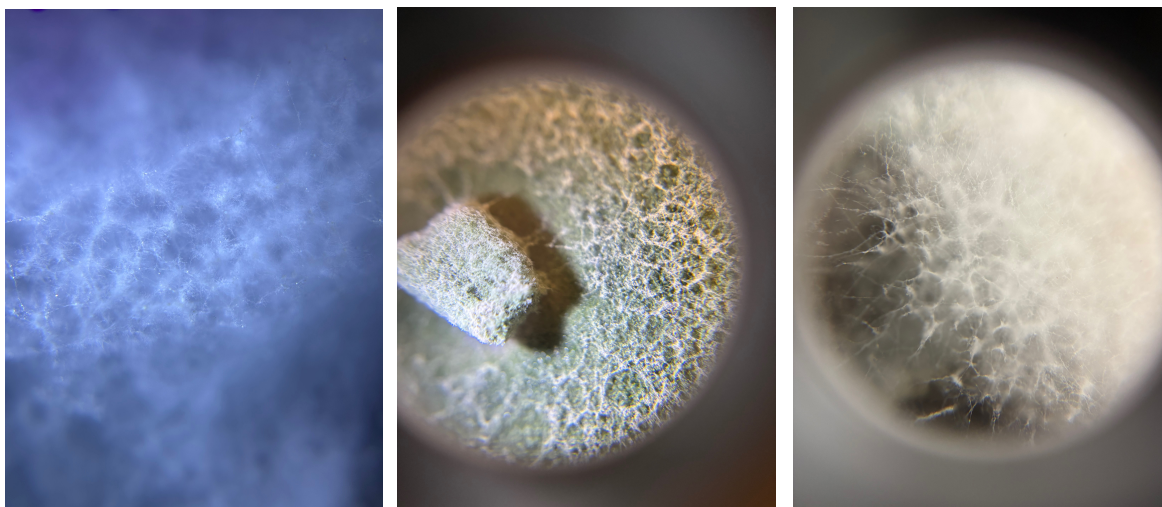
Appendix C.2 Orange



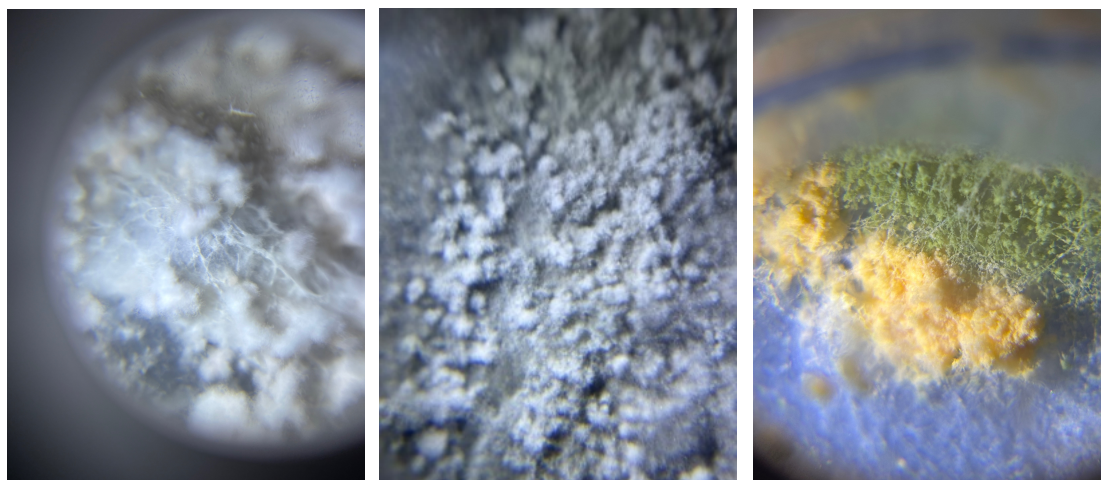
Appendix C.3 Green



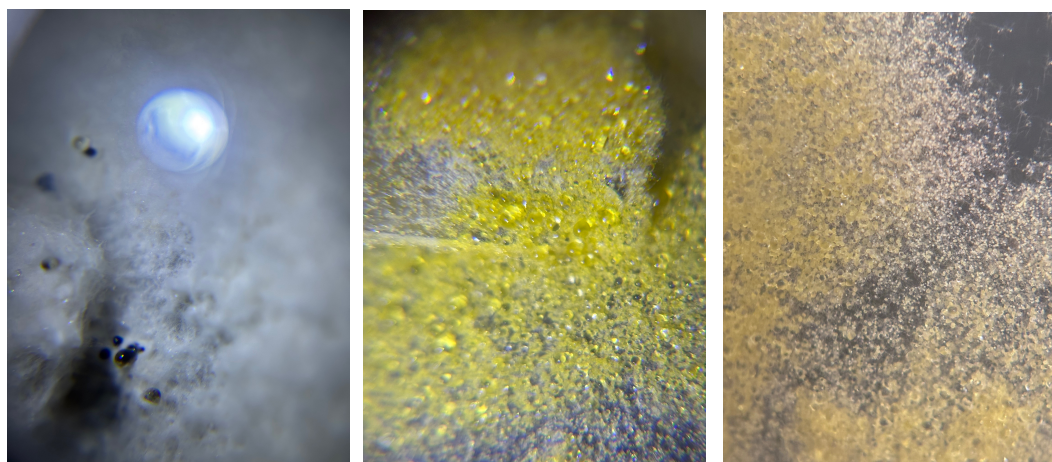
Appendix C.4 Gray



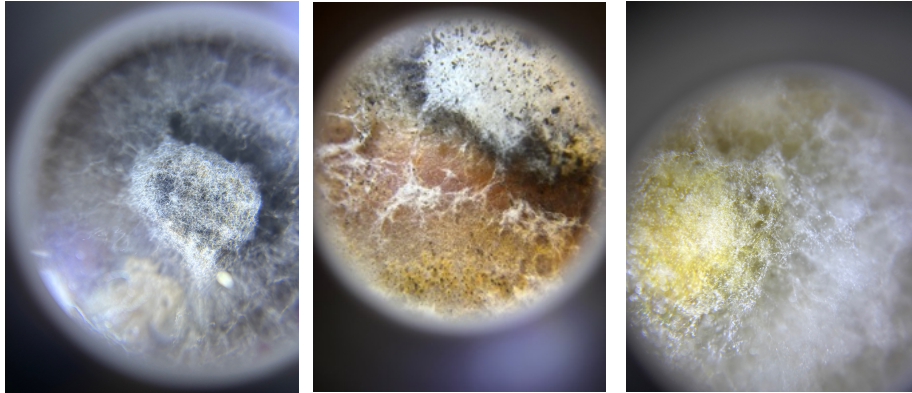
Appendix C.5 Geometric



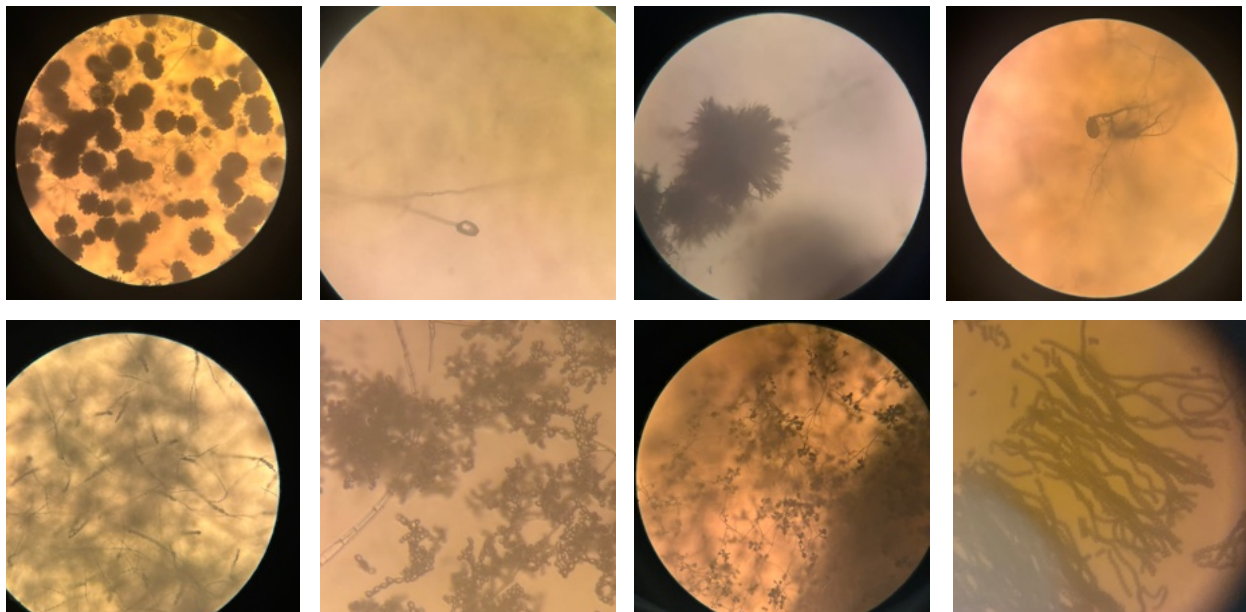
Appendix C.6 Cotton Ball



Appendix C.7 Bubble



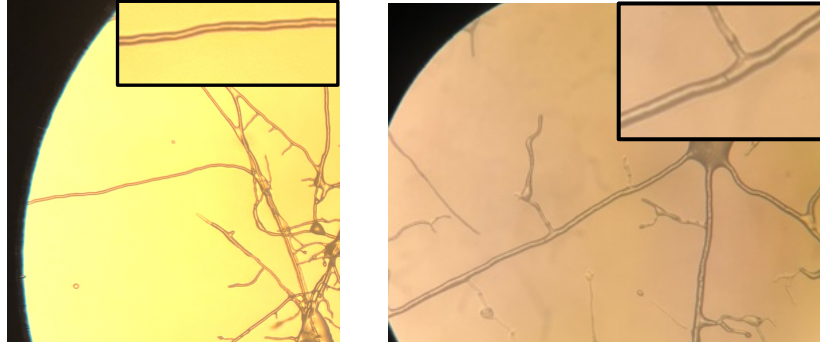
Appendix C.8 Spider Web



Appendix C.9

From Left to Right, Top to Bottom:

‘Blooming flowers, round, blooming flowers, bumpy,
elongated, clustered, floral clusters, horizontally clusters’



Appendix C.10 From Left to Right:

Septate, Aseptate

APPENDIX D

Labeling Key

TABLE 3.

Plate ID	Temperature Label (A, B, or C)	Unique Sample Number	Generation
A14.2	A	14	2
B52.1	B	52	1

APPENDIX E

Media Recipes

TABLE 4.

Media	Recipe
Potato Dextrose Agar (PDA)	200 g of washed but unpeeled potatoes slices were boiled in 1 L sterilized seawater (using a crock pot to autoclave for one heating cycle) for 30 min, 20 g sugar and 20 g agar powder was then added into the potato soup (Wang). Then, the mixture was heated in the microwave for 1 minute, cooled off for 1 minute, and repeated 5-6 times. Then the mixture was cooled for ten minutes before pouring into sterilized petri dishes.
Seawater Agar (SA)	1 L of sterilized seawater and 20 g agar powder mixed and heated in microwave following the same process as above.